Sequences and Series

Pure Mathematics Year 2

Chapter 3

Sequences and Series

Arithmetic sequences and series · geometric sequences and series · sigma notation · recurrence relations

Chapter Overview

1. Arithmetic series

Determine the value of

$2 + 4 + 6 + \cdots + 100$.

2. Geometric series

The first term of a geometric sequence is 3 and the second term is 1. Find its sum to infinity.

3. Sigma notation

Determine the value of

$$\sum_{r=1}^{100} (3r + 1).$$

4. Recurrence relations

If $a_1 = k$ and $a_{n+1} = 2a_n - 1$, determine a_3 in terms of k.

Types of sequences 3

1. $2, 5, 8, 11, 14, \ldots$

Name this type of sequence and describe the change between consecutive terms.

1

2. $3, 6, 12, 24, 48, \ldots$

Name this type of sequence and describe the change between consecutive terms.

2

3. $1, 1, 2, 3, 5, 8, \ldots$

How is each new term generated?

3

Definition

An arithmetic sequence has a constant difference between consecutive terms.

Geometric sequences and recurrence relations will be explored later in this chapter.

The fundamentals of sequences 4

Sequence notation

Consider

$2, 5, 8, 11, 14, \ldots$

What does n represent?

1

What does u_n represent?

2

For the value 8, state its position and its term notation.

3

The general term

Let a be the first term and d the common difference.

Moving to position n takes $n-1$ steps.

$u_1 = a,$

$u_2 = a + d,$

$u_3 = a + 2d.$

$$\boxed{u_n = a + (n-1)d}$$

Write down a, d and the relevant position before substituting.

Example 1

$u_n = 55 - 2n.$

(a) Find the first three terms.

(b) Find the first negative term.

1a

1b

Example 2

Find the general term of each sequence.

(a) $6, 20, 34, 48, 62, \ldots$

(b) $101, 94, 87, 80, 73, \ldots$

2a

2b

Further Examples 6

1. Find the constants

A sequence is generated by $u_n = an + b$, where a and b are constants. Given $u_3 = 5$ and $u_8 = 20$, find a and b.

1

2. Equal differences

For which values of x do -8, x^2 and $17x$ form the first three terms of an arithmetic sequence?

2

Edexcel C1 May 2014(R), Q10

Xin has a 14-day training schedule. She runs for A minutes on day 1 and increases her running time by $d + 1$ minutes each day, where A and d are constants.

(a) Show that on day 14 she runs for $A + 13d + 13$ minutes. (2 marks)

a

Yi also has a 14-day schedule. She runs for $A - 13$ minutes on day 1 and increases her running time by $2d - 1$ minutes each day.

(b) Yi and Xin run for the same time on day 14. Find d. (3 marks)

b

4. The first term of an arithmetic sequence is 14. The fourth term is 32. Find the common difference.

4

5. A sequence is generated by $u_n = pn + q$. Given $u_6 = 9$ and $u_9 = 11$, find p and q.

5

6. An arithmetic sequence has $u_3 = 30$ and $u_9 = 9$. Find its first negative term.

6

7. The 20th term is 14 and the 40th term is -6. Find the 10th term.

7

8. The first three terms are $5p$, 20 and $3p$. Find the 20th term.

8

10. An arithmetic sequence has first term k^2 and common difference k, where $k > 0$. The fifth term is 41. Find k, giving your answer in the form $p + q\sqrt{5}$, where p and q are integers. (4 marks)

10

STEP I 2004, Q5. The positive integers split into five arithmetic progressions:

$A : 1, 6, 11, 16, \ldots$ $B : 2, 7, 12, 17, \ldots$ $C : 3, 8, 13, 18, \ldots$

$D : 4, 9, 14, 19, \ldots$ $E : 5, 10, 15, 20, \ldots$

Give the general term of each progression. Prove that any term in B plus any term in C belongs to E.

1

Prove that the square of every term in B belongs to D. State and prove a similar claim for the square of every term in C.

2

Show there are no positive integer solutions:

(i) $x^2 + 5y = 243723$.

(ii) $x^4 + 2y^4 = 26081974$.

3

What is a series?

A series is a sum of terms in a sequence. In this syllabus, a series may be finite or infinite.

Arithmetic series

Add terms of an arithmetic sequence:

$2 + 5 + 8 + 11$.

Consecutive terms have a constant difference.

Taylor series

Functions can be expressed as infinite series of powers of x. For example:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This is studied in Further Mathematics.

Terminology

A power series is an infinite series of increasing non-negative integer powers of a variable, with constant coefficients. Power series also appear in Further Statistics.

Binomial series

A negative or fractional power can produce an infinite expansion. For $|x| < 1$:

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots$$

This is studied later in Year 2.

Partial sums

The harmonic series is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

A partial sum uses a finite number of terms:

$$H_3 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}.$$

The sum of an arithmetic series

Let a be the first term, d the common difference and S_n the sum of the first n terms.

Formula

Exam note

Be able to explain this proof. Symmetrically opposite terms add to the same total; the forward-and-reverse argument works for both odd and even n.

See the pairing

Consider $2, 5, 8, 11, 14, 17, \ldots$. Write the first five terms forwards and backwards.

$S_5 = 2 + 5 + 8 + 11 + 14,$

$S_5 = 14 + 11 + 8 + 5 + 2.$

Add corresponding terms.

Example

Prove the formula

Write the terms forwards and backwards, with $L = a + (n-1)d$. Each corresponding pair has the same sum.

Proof

Using the first and last terms

Suppose an arithmetic series has n terms, first term a and last term L:

$a + (a + d) + \cdots + L.$

Pair the two copies

Write the terms forwards and backwards. Each pair consists of terms equally far from opposite ends, so each pair sums to $a + L$.

There are n pairs in the two copies together.

Formula

Connection to the earlier formula

The last term is $L = a + (n-1)d$. Substitute this into the first-and-last-term formula.

Connection

Examples 15

Find the sum of the first 30 terms.

Write down a, d and n before substituting. Use S_n to distinguish a sum from the nth term.

1. $2 + 5 + 8 + 11 + 14 + \cdots$

1

2. $100 + 98 + 96 + \cdots$

2

3. $p + 2p + 3p + \cdots$

3

4. Find the minimum number of terms for $4 + 9 + 14 + \cdots$ to exceed 2000.

4

Test Your Understanding 16

Edexcel C1, January 2012, Q9

A company offers two salary schemes for Years 1 to 10 inclusive. Scheme 1 starts at $£P$, increasing by $£2T$ each year. Scheme 2 starts at $£(P + 1800)$, increasing by $£T$ each year.

(a) Show that the ten-year total under Scheme 1 is $£(10P + 90T)$. (2 marks)

a

(b) Both schemes give the same ten-year total. Find T. (4 marks)

b

(c) Scheme 2 pays £29,850 in Year 10. For this value of T, find P. (3 marks)

c

3. Find the sum of the first 50 even numbers.

3

4. Find the least number of terms for $7 + 12 + 17 + 22 + 27 + \cdots$ to exceed 1000.

4

5. The first term of an arithmetic series is 4. The sum to 20 terms is -15. Find, in any order, the common difference and the 20th term.

Find the common difference

Difference

Find the twentieth term

20th term

6. The sum of the first three terms of an arithmetic series is 12. The twentieth term is -32. Find the first term and common difference.

6

7. Prove that the sum of the first 50 natural numbers is 1275.

7

8. Show that the sum of the first $2n$ natural numbers is $n(2n+1)$.

8

9. Prove that the sum of the first n odd numbers is n^2.

9

10. The fifth term of an arithmetic series is 33 and the tenth term is 68. The sum of the first n terms is 2225.

(a) Show that $7n^2 + 3n - 4450 = 0$.

a

(b) Hence find n.

b

11. An arithmetic series is $(k + 1) + (2k + 3) + (3k + 5) + \cdots + 303$.

(a) Find the number of terms in terms of k.

a

(b) Show that the sum is $\frac{152k+46208}{k+2}$.

b

(c) Given $S_n = 2568$, find k.

c

1. MAT 2008, 1I

For positive integers n, let $S(n)$ be the sum of its digits; for example, $S(723) = 12$. Find

$S(1) + S(2) + \cdots + S(99)$.

1

2. AEA 2010, Q2

An arithmetic series satisfies $S_p = q$ and $S_q = p$, where p, q are positive integers and $p \neq q$.

In terms of p, q, find:

(a) the common difference;

(b) the first term;

(c) S_{p+q}.

Full solution on the next slide.

3. MAT 2007, 1J

For every $n \geq 1$,

$$(n+1) + (n^4 + 2) + \cdots + (n^{10000} + 100) > k.$$

Which follows?

A: $k < 1300$ B: $k^2 < 101$

C: $k \geq 101^{10000}$

D: $k < 5150$

3

AEA 2010, Q2

An arithmetic series has $S_p = q$ and $S_q = p$, where $p, q > 0$ are distinct integers. Find its common difference d, first term a and S_{p+q}.

(a) Common difference

a

(b) First term

b

(c) Sum to $p + q$ terms

c

Recap of Arithmetic vs Geometric Sequences 25

Compare the two sequences

Identify the type of sequence and its constant step.

1. $2, 5, 8, 11, 14, \ldots$

1

2. $3, 6, 12, 24, 48, \ldots$

2

Definition

A geometric sequence is one in which each term is obtained by multiplying the previous term by the same constant ratio.

Quickfire Common Ratio 26

Identify the common ratio r.

Divide a term by the preceding non-zero term.

1. $1, 2, 4, 8, 16, 32, \ldots$

1

2. $27, 18, 12, 8, \ldots$

2

3. $10, 5, 2.5, 1.25, \ldots$

3

4. $5, -5, 5, -5, 5, -5, \ldots$

4

5. $x, -2x^2, 4x^3, \ldots$

5

6. $1, p, p^2, p^3, \ldots$

6

7. $4, -1, 0.25, -0.0625, \ldots$

7

Alternating signs

A negative common ratio makes non-zero terms alternate between positive and negative. For item 5, if $x = 0$, all displayed terms are zero and the ratio cannot be uniquely determined.

Arithmetic sequence

Arithmetic formula

Geometric sequence

Geometric formula

1. Find the tenth and nth terms of $3, 6, 12, 24, \ldots$.

1

2. Find the tenth and nth terms of $40, -20, 10, -5, \ldots$.

2

Tip

Write down a and r before substituting. The exponent is $n - 1$, because there are $n - 1$ multiplications after the first term.

Textbook example

The second term of a geometric sequence is 4 and the fourth term is 8. Its common ratio is positive. Find the exact values of the common ratio, the first term and the tenth term.

(a) Common ratio

a

(b) First term

b

(c) Tenth term

c

Tip

Writing $u_2 = 4$ first helps distinguish the second term from the sum of the first two terms.

Further Example 29

The first three terms of a geometric sequence are $3, x, x + 6$. All terms are positive.

Find x and the tenth term.

1. Find x

1

2. Find the tenth term

2

Method

Equate consecutive ratios when three consecutive terms are given. Check which solutions satisfy the conditions in the question.

nth term with inequalities 30

Find the first term of the geometric sequence $3, 6, 12, 24, \ldots$ that exceeds one million.

1. Find its position

1

2. Calculate and check the term

2

Tip

The position n must be an integer. Use the inequality to choose the first valid position, then calculate the term.

Test Your Understanding 31

1. A geometric sequence has only positive terms. Its third term is 20 and its fifth term is 80. Find the twentieth term.

1

2. The second, third and fourth terms of a geometric sequence are $x, x+6, 5x-6$.

(a) Find the possible values of x.

(b) Find r, given that it is positive.

(c) Hence find the first term.

2a

2b–c

Exercise 3C , Year 2 Pages 69-70 32

Exercise 3C, question 3

If $3, x, 9$ are the first three terms of a geometric sequence, find:

(a) the exact value(s) of x;

(b) the exact value(s) of the fourth term.

(a) Second term

a

(b) Fourth term

b

Sign check

No positive-ratio condition is stated. Remember to consider both square roots when solving an equation of the form $x^2 = k$, where $k > 0$.

5. The nth term of a geometric sequence is 2×5^n. Find the first and fifth terms.

5

6. The sixth term of a geometric sequence is 32 and the third term is 4. Find the first term and the common ratio.

6

8. The first three terms of a geometric sequence are $8 - x, 2x, x^2$, where $x > 0$.

(a) Show that $x^3 - 4x^2 = 0$. (2 marks)

a

(b) Find the twentieth term. (3 marks)

b

(c) Is 4096 a term of the sequence? Give a reason. (1 mark)

c

Exercise 3C , Year 2 Pages 69-70 35

9. A geometric sequence has first term 200 and common ratio $p > 0$. Its sixth term is 40.

(a) Show that $5 \log p + \log 5 = 0$.

(b) Find p to 3 significant figures.

9

11. The first three terms of a geometric sequence are $9, 36, 144$. State, with a reason, whether 383616 is a term of the sequence.

11

Sum of the first n terms of a geometric series 36

Arithmetic series

Arithmetic formula

Geometric series

Geometric formula

Proof of the geometric-series formula

Write the sum, multiply by r, then subtract.

Proof

Examples 37

Find the sum of the first ten terms of each geometric series.

Use $S_n = \dfrac{a(1 - r^n)}{1 - r}$ for $r \neq 1$.

1. $3 + 6 + 12 + 24 + 48 + \cdots$

1

2. $4 + 2 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

2

Check

A term u_n and a sum S_n are different quantities. Identify which one the question asks for before choosing a formula.

Harder Example 38

Find the least value of n such that the sum of $1 + 2 + 4 + 8 + \cdots$ to n terms exceeds $2\,000\,000$.

1. Form and solve an inequality

1

2. Verify the boundary

2

Why logarithms?

Taking logarithms lets us solve for an unknown exponent. Since the base 2 exceeds 1, the inequality keeps its direction.